$A\vec{x} = b$ $\begin{bmatrix}1&1\\2&1\\3&1\end{bmatrix}\begin{bmatrix}m\\b\end{bmatrix} = \begin{bmatrix}y_1\\2\\2\end{bmatrix}$ $(1,1)(2,2)(3,2)$

$A^TA\vec{x} = A^Tb$ $m\cdot 1 + b = 1$

$\vec{x} = (A^TA)^{-1}A^Tb$ $m\cdot 2 + b = 2$

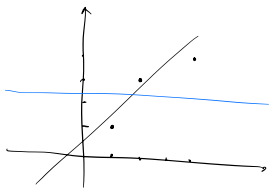
$m\cdot 3 + b = 2$

data

$Xb = Y$

similarly $b = (X^TX)^{-1}X^TY = \frac{1}{n\sum(x_i-\bar{x})^2}\begin{bmatrix}\sum x_i^2 & -\sum x_i\\ -\sum x_i & n\end{bmatrix}\begin{bmatrix}\sum y_i\\ \sum x_i y_i\end{bmatrix}$

$= \begin{pmatrix}\bar{y} - \bar{x}\frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2}\\ \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2}\end{pmatrix} = \begin{bmatrix}b_0\\ b_1\end{bmatrix}$



prove: $\hat{\beta}_1 = r_{xy}\frac{s_y}{s_x}$

$\vec{y} = \beta_0\mathbf{1} + \beta_1 x + \varepsilon$ $H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T$ centering matrix

$x_c := Hx = x - \bar{x}\mathbf{1}$

$y_c := Hy = y - \bar{y}\mathbf{1}$

$\min_b \|y_c - bx_c\|^2$

By orthogonality $x_c^T(y_c - \hat{\beta}_1 x) = 0$

$x_c^T y_c = x_c^T\hat{\beta}_1 x$

$\hat{\beta}_1 = \frac{x_c^T y_c}{x_c^T x_c} = \frac{\|x_c\|\|y_c\|\cdot\cos\theta}{\|x_c\|^2}$

note $r_{xy} = \frac{x_c^T y_c}{\|x_c\|\|y_c\|} = \cos\theta$

$\cos\theta\frac{\|y_c\|}{\|x_c\|} = r_{xy}\frac{\|y_c\|}{\|x_c\|}$ $s_x = \frac{\|x_c\|}{\sqrt{n-1}}$

$= r_{xy}\frac{s_y}{s_x}$ $s_y = \frac{\|y_c\|}{\sqrt{n-1}}$

$SE(\hat{\beta}_1) = \frac{\hat{\sigma}}{\|x_c\|}$

prove: $e = y_c - \hat{\beta}_1 x_c$

$\hat{\beta}_1 x_c + e = y_c$

$\|\hat{\beta}_1\|^2\|x_c\|^2 + e^2 = \|y_c\|^2$

previous $\hat{\beta}_1 = r_{xy}\frac{\|y_c\|}{\|x_c\|}$

$\Rightarrow e^2 = (1 - r_{xy}^2)\|y_c\|^2$

$\hat{\sigma}^2 = \frac{\|e\|^2}{n-2}$ residual variance

$SE(\hat{\beta}_1) = \frac{\hat{\sigma}}{s_x\sqrt{n-1}} = \frac{\sqrt{(1-r_{xy}^2)\|y_c\|^2/(n-2)}}{\|x_c\|/\sqrt{n-1}} = \frac{s_y}{s_x}\sqrt{\frac{1-r_{xy}^2}{n-2}}$

$r_{xy}\cos\theta = r_{xy}\frac{\|x_c\|}{\|y_c\|} = r_{xy}\frac{s_x\sqrt{n-1}}{s_y\sqrt{n-1}}$

$= \cot\theta\sqrt{n-2}$

$Y = \begin{pmatrix}Y_1\\Y_2\\Y_3\end{pmatrix}$ $\mathbf{1} = \begin{pmatrix}1\\1\\1\end{pmatrix}$ $X = \begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}$ $\varepsilon\begin{pmatrix}\varepsilon_1\\\varepsilon_2\\\varepsilon_3\end{pmatrix}$

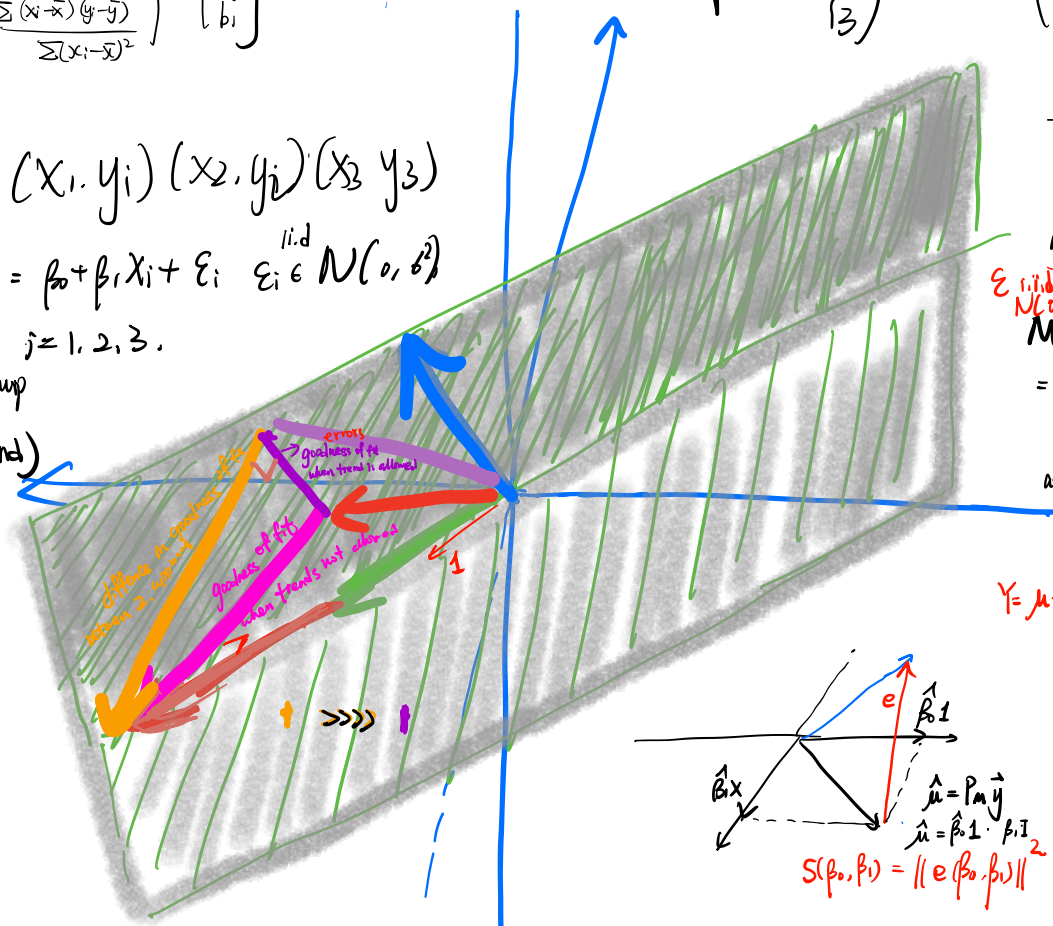
$(x_1, y_1)(x_2, y_2)(x_3, y_3)$

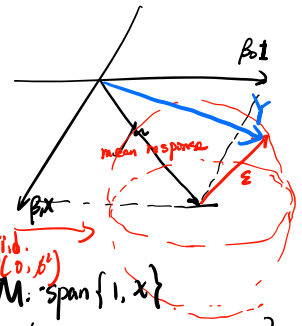
$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ $\varepsilon_i \overset{i.i.d}{\sim} N(0, \sigma^2)$

$j = 1, 2, 3.$

i) no additional assump

ii) $\beta_1 = 0$ (no trend)





$\varepsilon \overset{i.i.d}{\sim} N(0, \sigma^2)$

$M = \text{span}\{\mathbf{1}, x\}$

$= \{\beta_0\mathbf{1} + \beta_1 x : \beta_0, \beta_1 \in \mathbb{R}\}$

$\subseteq \mathbb{R}^2 \subseteq \mathbb{R}^3$

assuming x & $\mathbf{1}$ lin. ind.

$(x_1 = x_2 = x_3)$ not applied.

$Y = \mu + \varepsilon$ $\mu \in M$. $\varepsilon \in N(0, \sigma^2 I_3)$ identity for $\mathbb{R}^3$

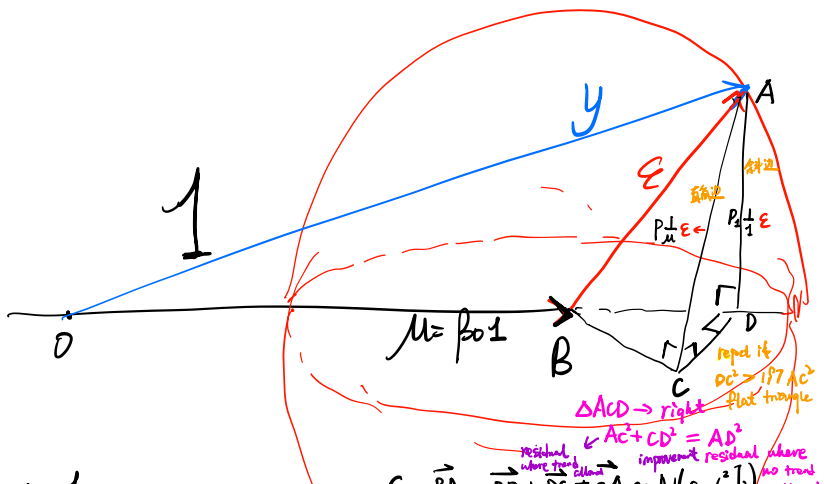
$S(\beta_0, \beta_1) = \|e(\beta_0, \beta_1)\|^2$

Therefore $H_0: \mu \in \mathbf{1}$

$H_1: \mu \in M$

from D (data) D measure deviation.

from H_0 D should follow a parameter-free distribution under H_0



$\varepsilon = \vec{BA} = \vec{BD} + \vec{DC} + \vec{CA} \sim N(0, \sigma^2 I_3)$

if they're mutual orthogonal

$\vec{BD}, \vec{DC}, \vec{CA}$ $i.i.d$ $N(0, \sigma^2)$

obvious

prove. $\vec{BD} \perp \vec{DC}$

$\vec{DC} = \vec{AC} - \vec{AD}$

$\vec{BD}\cdot\vec{DC} = \vec{BD}(\vec{AC} - \vec{AD})$ ✓

ANOVA Table	Sum of Squares	DoF	Distribution
Goodness of fit:	$\|P_{\mathbf{1}}^{\perp}Y\|^2$	2	$\sigma^2\chi^2_{(2)}$
Reduced model	$\|P_M^{\perp}Y\|^2$	1	$\sigma^2\chi^2_{(1)}$
Difference	$\|P_{\mathbf{1}}^{\perp}Y - P_M^{\perp}Y\|^2$	1	$\sigma^2\chi^2_{(1)}$

Under $H_0: \mu \in \mathbf{1}$

$F := \frac{\|P_{\mathbf{1}}^{\perp}Y\|^2 - \|P_M^{\perp}Y\|^2}{\|P_M^{\perp}Y\|^2} = \frac{DC^2}{AC^2} \overset{d}{=} \frac{U^2}{V^2}$

U, V $i.i.d$ $N(0,1)$

$Y \sim$ Cauchy

Let $X \sim$ Cauchy

The distribution of a squared standard Cauchy is $F_{1,1}$.

define $Y = X^2$.

$$P(X \le x) = \frac{1}{\pi}\arctan x + \frac{1}{2}$$

for $y \ge 0$

$$P(Y \le y) = P(X^2 \le y) = P(-\sqrt{y} \le x \le \sqrt{y})$$
$$= \frac{1}{\pi}\arctan\sqrt{y} + \frac{1}{2} - \frac{1}{\pi}\arctan(-\sqrt{y}) - \frac{1}{2}$$
$$= \frac{2}{\pi}\arctan(y) = p$$

Set $P(Y \le y) = p$ $\quad p = \frac{2}{\pi}\arctan\sqrt{y}$

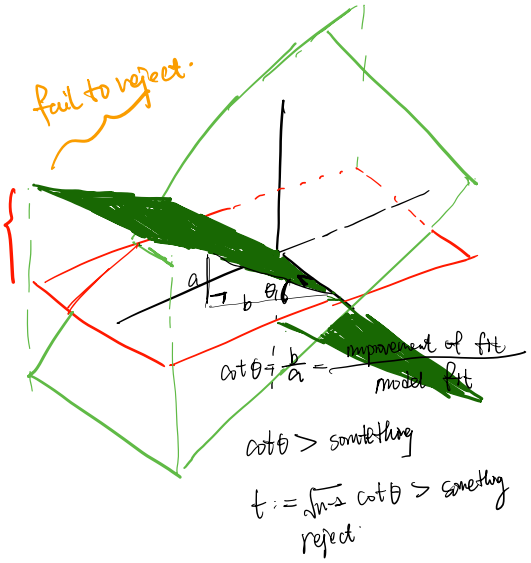
$\rightarrow y = \tan^2(\pi p/2)$ critical value

for $0 < p < 1$ $\quad 100p\%$ quantile $F_{1,1}$ is $\tan^2(\pi p/2)$

95%, $p = 0.95$ $\quad F_{1,1} = \tan^2\left(\frac{2\pi \cdot 0.95}{2}\right) = 161.45$

Reject H_0 if $\quad F := \frac{\|P_+ Y\|^2 - \|P_{\mathbb{1}} Y\|^2}{\|P_{\mathbb{1}}^\perp Y\|^2} > F_{1,1,\alpha} = \tan^2\left(\frac{\pi(1-\alpha)}{2}\right)$

significance level $\alpha \in (0,1)$



Hypothesis Test: Is the observe data too far from model constrained by b

CI: which set of b give a model close enough to the data

construct CI: $H_0: \beta_1 = 0$

$\rightarrow H_0: \beta_1 = b$

$\mathcal{L}_b = \{\beta_0 \mathbb{1} + bX,\ b \in \mathbb{R}\}$

$$F := \frac{\|C - D_b\|^2}{\|Y - C\|^2}$$

C: full model projection point.

D_b: projection on to the new sub-plane point.

Centered predictor vector

$X_c = X - \bar{x}\mathbb{1}$

$\mathbb{1}^T X_c = 0$ $\quad \mathbb{1}, X_c$ orthogonal

$C = \bar{y}\mathbb{1} + \hat{\beta}_1 X_c$ mean value of y + $\hat{\beta}$ · movement of x

$D_b = \bar{y}\mathbb{1} + bX_c$ $\quad b = 0$ in special case

$C - D_b = (\hat{\beta}_1 - b)X_c$

$(C - D_b)^2 = (\hat{\beta}_1 - b)^2 \|X_c\|^2$ Directly OD

$= (\hat{\beta}_1 - b)^2 \sum_{i=1}^n (x_i - \bar{x})^2 = S_{xx}$

$$\rightarrow F: \frac{(\hat{\beta}_1 - b)^2 S_{xx}}{\|Y - C\|^2/(n-2)} = \frac{(\hat{\beta}_1 - b)^2 S_{xx}}{s^2}$$

residual variance

$F(b) \le F_{1,n-2;0.95}$ 95% percentile of F-distribution

$$\rightarrow \frac{(\hat{\beta}_1 - b)^2 S_{xx}}{s^2} \le F_{\cdot\cdot}$$

$$|\hat{\beta}_1 - b| \le \sqrt{F_{1,n-2;0.95}} \cdot \frac{s}{\sqrt{S_{xx}}}$$

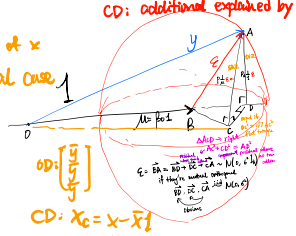
$$b \in \left[\hat{\beta}_1 \pm \sqrt{F_{1,n-2;0.95}} \frac{s}{\sqrt{S_{xx}}}\right]$$

This approach directly calculate CD

AC: $Y - C$ Residual for full model

AD: $Y - D$ residual for intercept only mode

CD: additional explained by X



OD: $\begin{bmatrix} \bar{y} \\ \bar{y} \end{bmatrix}$

CD: $X_c = X - \bar{x}\mathbb{1}$

$= \vec{X} - \begin{bmatrix} \bar{x} \\ \bar{x} \end{bmatrix}$

OD: $\bar{y}\mathbb{1}$

OC: $\bar{y}\mathbb{1} + \hat{\beta}\left(X - \begin{bmatrix} \bar{x} \\ \bar{x} \end{bmatrix}\right)$

Special. $n - 2 = 1$

$F(b) \sim F_{1,1}$

$P(F \le f) = \frac{1}{\pi}\arctan\sqrt{f} = 0.95$

$f^* = \tan^2\left(\frac{0.95\pi}{2}\right) = 161.45$

$\rightarrow \hat{\beta}_1 \pm \sqrt{161.45}\frac{s}{\sqrt{S_{xx}}}$

Under H_0: $\mu \in \mathbb{1}$

$$F := \frac{\|P_{\mathbb{1}}^\perp Y\|^2 - \|P_{\mathcal{M}}^\perp Y\|^2}{\|P_{\mathbb{1}}^\perp Y\|^2} = \frac{DC^2}{AC^2} \overset{d}{=} \frac{U^2}{V^2}$$

U, V i.i.d $N(0,1)$

AD

$P_1 y - y \perp \mathbb{1}$ $\quad P_1 y = c\mathbb{1}$

$\rightarrow (P_1 y - y)^T \mathbb{1} = 0$

$(c\mathbb{1} - y)^T \mathbb{1} = 0$

$(c\mathbb{1})^T \mathbb{1} - y^T \mathbb{1} = 0$

$$c = \frac{y^T \mathbb{1}}{\mathbb{1}^T \mathbb{1}} = \frac{y_1 + y_2 + y_3}{(1,1,1)(1,1,1)} = \bar{y}$$

$P_1^\perp y = (y_1 - \bar{y}, y_2 - \bar{y}, y_3 - \bar{y})^T$ (no trend model error)

Let $e_1 = \frac{\mathbb{1}}{\|\mathbb{1}\|} = \begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}$

$$e_2 = \frac{P_1^\perp X}{\|P_1^\perp X\|} = \frac{(x_1 - \bar{x}, x_2 - \bar{x}, x_3 - \bar{x})}{l \qquad l^2}$$

$\hat{\mu} = P_{\mathcal{M}} y = (y^T e_1)e_1 + (y^T e_2)e_2$ prediction

$P_{\mathcal{M}}^\perp y = y - \hat{\mu}$ (trend model error)

difference = additional residual explained.

OA: $\hat{\mu}$ $\quad AD^2$

OD: y $\quad AC^2$

$$F: \frac{\|(y_1 - \bar{y}, y_2 - \bar{y}, y_3 - \bar{y})\|^2 - \|(y - \hat{\mu})\|^2}{\|(y - \hat{\mu})\|^2} \quad y^T \frac{1}{m} y$$

residual of null model. $\quad$ residual of full

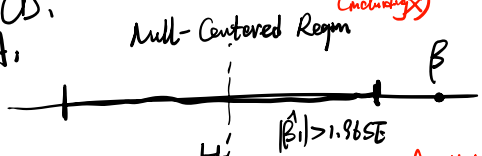
pythagorean for OD

This approach use

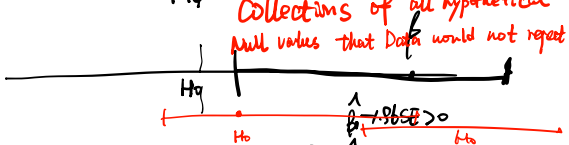
residual square for the full model.

Pythagorean for CI:

A:



B.



$\Leftrightarrow |\hat{\beta}_1| > 1.96SE$

Collections of all hypothetical Null values that Data would not reject

distribution of estimator under Null

CI: plausible population parameter given the observed estimate.

$$\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} \qquad \hat{\beta} \sim N(\beta, \sigma^2 (X^T X)^{-1})$$

reason why $\hat{\beta}$ follows normal

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$
$$Y = X\beta + \varepsilon$$
$$= (X^T X)^{-1} X^T (X\beta + \varepsilon)$$
$$= (X^T X)^{-1} X^T X \beta + (X^T X)^{-1} X^T \varepsilon$$
$$= \beta + (X^T X)^{-1} X^T \varepsilon \rightarrow \text{also normal}$$

$$\hat{\beta}_j \sim N(\beta_j, \sigma^2 [(X^T X)^{-1}]_{jj})$$

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2}{n-p} = \frac{RSS}{n-p}$$

Therefore estimated

$$\widehat{SE}(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 [(X^T X)^{-1}]_{jj}}$$
$$\approx \sqrt{\hat{\sigma}^2 [(X^T X)^{-1}]_{jj}}$$

$$T = \frac{\hat{\beta}_j - \beta_j}{\widehat{SE}(\hat{\beta}_j)} \sim t_{n-p}$$

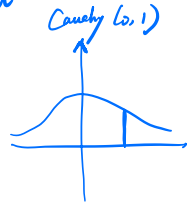
Note Cauchy

$t_1 = \text{Cauchy}(0,1)$

$$p = 2P(T_n \geq |t_{obs}|)$$

what's the prob that under the null we get as extreme as our data.

Cauchy (0, 1)



95% $P(-t_{0.975,\mu} \leq T \leq t_{0.975,\nu}) = 0.95$

$\Leftrightarrow P\left(-t_{0.975,\mu} \leq \frac{\hat{\beta}_j - \beta}{\widehat{SE}(\hat{\beta}_j)} \leq t_{0.975,\nu}\right) = 0.95$ ($\beta = 0$)

-1.96 $\quad +1.96$ (non-necessary, might be bigger than 1.96.)

$\Leftrightarrow \Leftrightarrow \left|\frac{\hat{\beta}_j}{\widehat{SE}}\right|$

$\Leftrightarrow \hat{\beta}_j - \beta \in [-1.96\widehat{SE}, 1.96\widehat{SE}]$

$\Leftrightarrow \beta - \hat{\beta}_j \in [-1.96\widehat{SE}, 1.96\widehat{SE}]$

$\Leftrightarrow \beta \in [\hat{\beta}_j - 1.96\widehat{SE}, \hat{\beta}_j + 1.96\widehat{SE}]$

p-2side	95% CI	Interpret
$p<0.05$	Excludes Zero	Reject $H_0=0$
$p>0.05$	Contains 0.	Fail to reject $H_0=0$

If there's single regression coefficient: test statistic t.

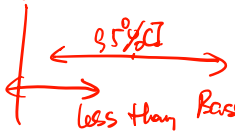
If there's multiple coef. F-statistic

$H_0: \beta_1 = \beta_2 = \beta_3 = 0$.

$F_{1,n-p} = T_{n-p}^2$ special case

$\beta = 0 \notin [\hat{\beta}_j - 1.96\widehat{SE}, \hat{\beta}_j + 1.96\widehat{SE}]$

$\Leftrightarrow |\hat{\beta}_j| > t_{0.975,n-p}\widehat{SE}$



less than 5% of chance that $H_0=0$

Based on data

	Sample Mean.	Regression Coef.
Estimator	$\bar{X}$	$\hat{\beta}_j$
true	μ	β_j
SE	$S/\sqrt{n}$	$\sqrt{\hat{\sigma}^2[(X^TX)^{-1}]_{jj}}$ where $\hat{\sigma}^2 = \frac{RSS}{n-p} = \frac{\sum_{i=1}^{n}(Y_i - \hat{\beta}_j)^2}{n-p}$
95%	$\bar{X} \pm t^* SE$	$\hat{\beta}_j \pm t^* SE$

95% CI: if we repeatedly sample data & construct interval using this procedure 95% of those intervals would contain the true coefficient.

$\Leftrightarrow$

$\not\Leftarrow$ After observing data, 95% probability that the fix parameter was in CI.

when there's only 1 predictor & intercept

$$X^T X \begin{bmatrix} X_1 - \bar{X}, X_2 - \bar{X}, \cdots X_n - \bar{X} \end{bmatrix} \begin{bmatrix} X_1 - \bar{X} \\ \vdots \\ \end{bmatrix}$$

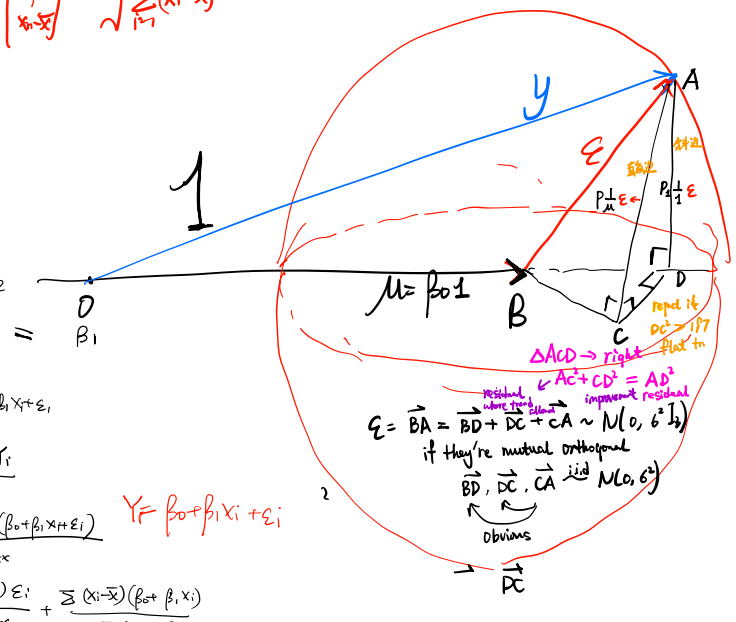
$$\widehat{SE}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}} = \sqrt{\hat{\sigma}^2 [(X^TX)^{-1}]_{11}}$$

$F = t^2$ one regression coef

$$F = \frac{(RSS_0 - RSS_1)/q}{RSS/n-p}$$

$$= \frac{\hat{\beta}_1^2 S_{xx}}{s^2} = \left(\frac{\beta_1}{s/\sqrt{S_{xx}}}\right)^2 = \left(\frac{\beta_1}{SE(\beta_1)}\right)^2 = t^2$$

model sd.



$$S_{xx} = \sum_i (x_i - \bar{x})^2 > 0 \qquad Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{S_{xx}} = \frac{\sum (x_i - \bar{x}) Y_i}{S_{xx}}$$

$$= \frac{\sum (x_i - \bar{x})(\beta_0 + \beta_1 x_i + \varepsilon_i)}{S_{xx}} \qquad Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$= \frac{\sum (x_i - \bar{x}) \varepsilon_i}{S_{xx}} + \frac{\sum (x_i - \bar{x})(\beta_0 + \beta_1 x_i)}{\sum (x_i - \bar{x})^2}$$

$$= \frac{\sum (x_i - \bar{x}) \varepsilon_i}{S_{xx}} + \frac{\beta_0 \sum (x_i - \bar{x})}{\sum (x_i - \bar{x})^2} + \frac{\beta_1 \sum (x_i - \bar{x}) x_i}{S_{xx}}$$

$$= \frac{\sum (x_i - \bar{x}) \varepsilon_i}{S_{xx}} + \frac{\beta_1 \sum (x_i - \bar{x})(x_i - \bar{x})}{S_{xx}} + \frac{\bar{x} \sum (x_i - \bar{x})}{S_{xx}}$$

$$= \beta_1 + \frac{\sum (x_i - \bar{x}) \varepsilon_i}{S_{xx}}$$

$$Var(\hat{\beta}_1) = \frac{1}{S_{xx}^2} Var\left(\sum (x_i - \bar{x}) \varepsilon\right) \qquad \text{note } Y_i = \beta_0 + \beta_1 X + \varepsilon$$

$$= \frac{1}{S_{xx}^2} \sum Var\left[(x_i - \bar{x}) \varepsilon\right] \qquad \varepsilon \text{ here means model sd error}^2$$

$$= \frac{1}{S_{xx}^2} (x_i - \bar{x})^2 \, Var(\varepsilon)$$

$$= \frac{1}{S_{xx}^2} \sum (x_i - \bar{x})^2 \cdot \sigma^2 = \frac{\sigma^2 S_{xx}}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}}$$

$$SE(\hat{\beta}_1) = \frac{\sigma}{\sqrt{S_{xx}}} \qquad \text{model sd error}$$

$$\sqrt{\hat{\sigma}^2 \left[(X^T X)^{-1}\right]_{jj}} \qquad \text{where } \hat{\sigma} = \frac{RSS}{n-p}$$

univariate case

$$\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} \to \frac{\hat{\sigma}}{\sqrt{S_{xx}}}$$

$$F = \frac{(RSS_0 - RSS)}{RSS/n-p}$$

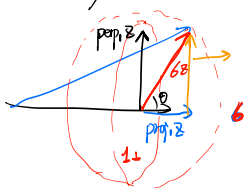
$$= \frac{\hat{\beta}_1^2 S_{xx}}{s^2} = \left(\frac{\beta_1}{s/\sqrt{S_{xx}}}\right)^2 = \left(\frac{\beta_1}{SE(\beta_1)}\right)^2 = t^2$$

model sd.

$X_1, X_2, \cdots X_n \overset{iid}{\sim} N(\mu, \sigma^2)$

$\bar{X}_n := \frac{\sum X_k}{n} \perp S^2 := \frac{1}{n-1}\sum_{k=1}^{n}(X_k - \bar{X}_n)^2$

$\sim N(\mu, \frac{\sigma^2}{n})$ $\quad \sim \frac{\sigma^2}{n-1}\chi^2_{(n-1)}$



proj₁ Z ⊥ perp₁ Z — orthogonal & independent

$z' \cdot 1 = \bar{z}\sqrt{n}$ Using: the coordinates of Gaussian relative to any orthonormal bases are iid normal

Therefore $\|\text{perp}_1 z\|^2 \sim \chi^2_{(n-1)}$

Sum of Squares	DF	Distribution
$\|X-\mu\mathbf{1}\|^2$	n	$\sigma^2\chi^2_{(n)}$
$\|\text{proj}_1 X-\mu\mathbf{1}\|^2$	1	$\sigma\chi^2_{(1)}$
$\|\text{perp}_1 x\|^2$	$n-1$	$\sigma\chi^2_{(n-1)}$

Under normal distribution: True Too / Gossip Theorem

sample mean ⊥ sample variance

$T = \frac{\sqrt{n}(\bar{X}-\mu)}{S}$ (sample sd) $= \frac{\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)S^2}{\sigma^2}\frac{1}{(n-1)}}}$ $\to Z \sim N(0,1)$

$Q \sim \chi^2_V$ $V = n-1$

$= \sqrt{n-1}\cdot\frac{\text{scalar. proj}_1 z}{\|\text{perp}_1 z\|} \sim$ $N(0,1)$ / $\chi^2_{(n-1)}$ / Q

$= \sqrt{n-1}\cot\theta$

Therefore $\sqrt{V}\cot\theta \sim t_V$

Recall 2 Def'n of t-statistic

Def'n 1: $T = \frac{Z}{\sqrt{W/V}}$ $Z \sim N(0,1)$, $W \sim \chi^2_{(V)}$

follows a t-distribution with $\sqrt{}$ degree of freedom

Def'n 2: $f_{t(V)} = \frac{\Gamma(\frac{V+1}{2})}{\sqrt{V\pi}\,\Gamma(\frac{V}{2})}\left(1+\frac{t^2}{V}\right)^{-\frac{V+1}{2}}$

$-\infty < t < +\infty$

Sample proportions

$X \sim Ber(p)$ $\mu(X) = p$, $Var(X) = \sqrt{p(1-p)}$

$X_1 + X_2 + \cdots + X_n \sim$ Binomial (p) $\mu(X) = np$ $Var(nX) = \sqrt{np(1-p)}$

$X + X + X \cdots$ By CLT

$\bar{X} \sim$ Roughly normal $\mu(\bar{X}) = n$ $\sigma(\bar{X}) = \sqrt{\frac{p(1-p)}{n}}$

$\rightarrow \frac{X_1 + X_2 + \cdots + X_n}{n}$

CI: $\hat{p} \pm z^* \sigma(\bar{X})$

$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

If true p_0, minimal size?

$|\hat{p} - p_0| > z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

$n > (\frac{z^*}{p - p_0})^2 \hat{p}(1-\hat{p})$

Two Sample t- statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n} + \frac{s_2^2}{n}}}$$

CI: $(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{s_1^2}{n} + \frac{s_2^2}{n}}$

Two - Sample Proportions

$SE = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$

CI: $(\hat{p}_1 - p_1) \pm z^* SE$.